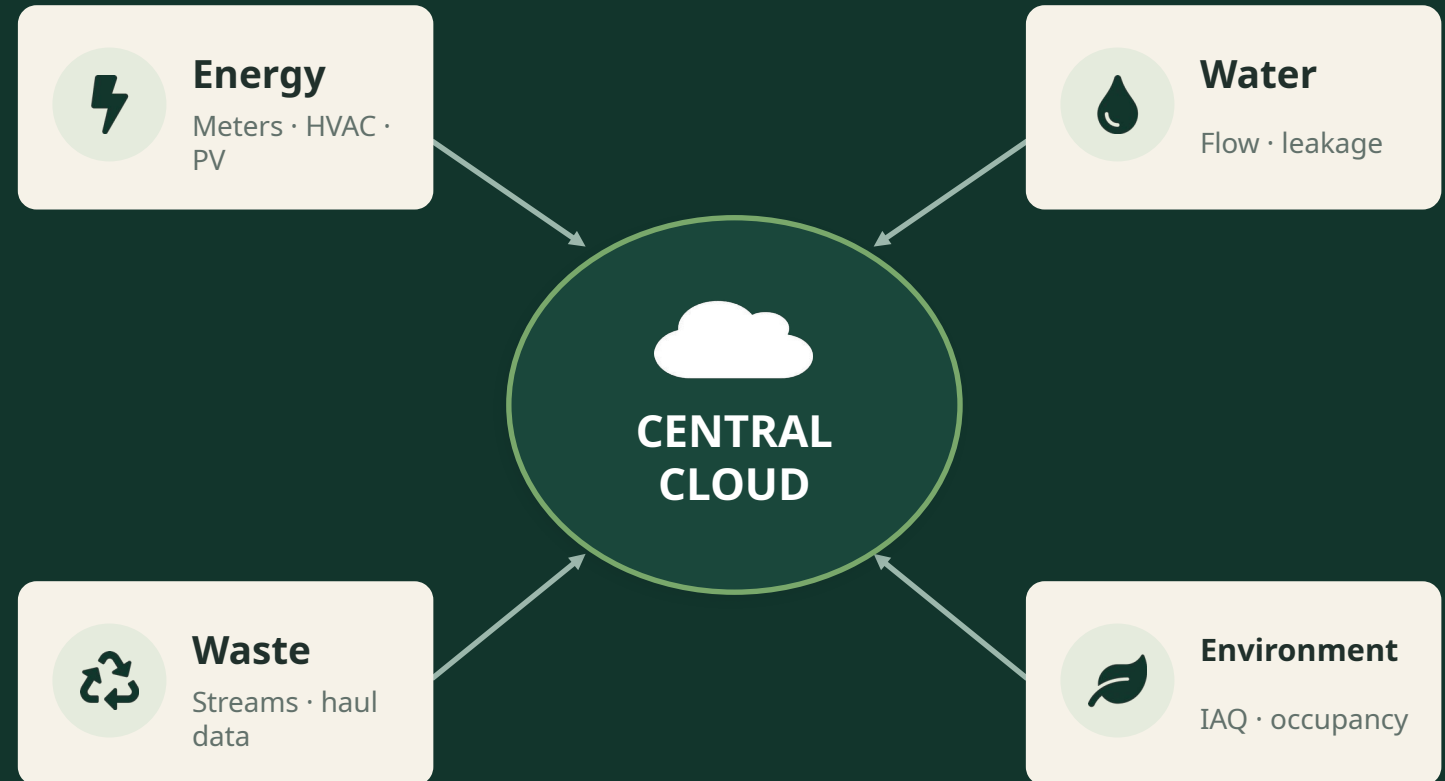


Visibility without local intelligence

Current Net Zero platforms centralize data, but decisions remain distant from the assets that consume resources.

-  **Latency** Response lags behind changing conditions
-  **Connectivity** WAN outage interrupts analytics
-  **Privacy** Fine-grained data leave the site
-  **Control gap** Dashboards report, not optimize



Consequence High visibility, limited real-time optimization and localized decision-making.

MOTIVATION

Move decisions closer to physical assets

Edge intelligence is valuable only when it improves response, resilience and evidence—not merely because inference is local.



< 3 min

EXECUTION BUDGET

Complete each cycle within 20% of the 15-minute control interval.



On-device

PRIVACY BOUNDARY

Convert occupancy modalities into aggregate features before external transfer.



24 h

WAN-OUTAGE OBJECTIVE

Continue approved local analytics with freshness-gated cached inputs.

CENTRALIZED

raw streams



EDGE

events + KPIs



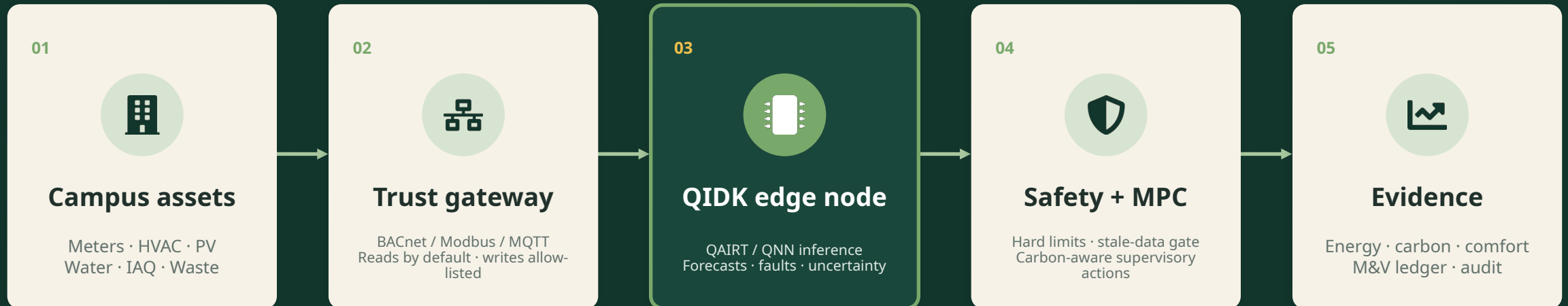
filter + aggregate locally

Less traffic • Faster bounded decisions • Measured overhead

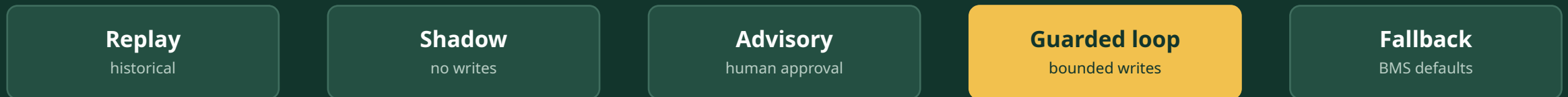
PROPOSED SOLUTION

A bounded edge-AI control plane

Qualcomm QIDK hosts prediction and supervisory optimization; the existing BMS retains equipment and life-safety authority.



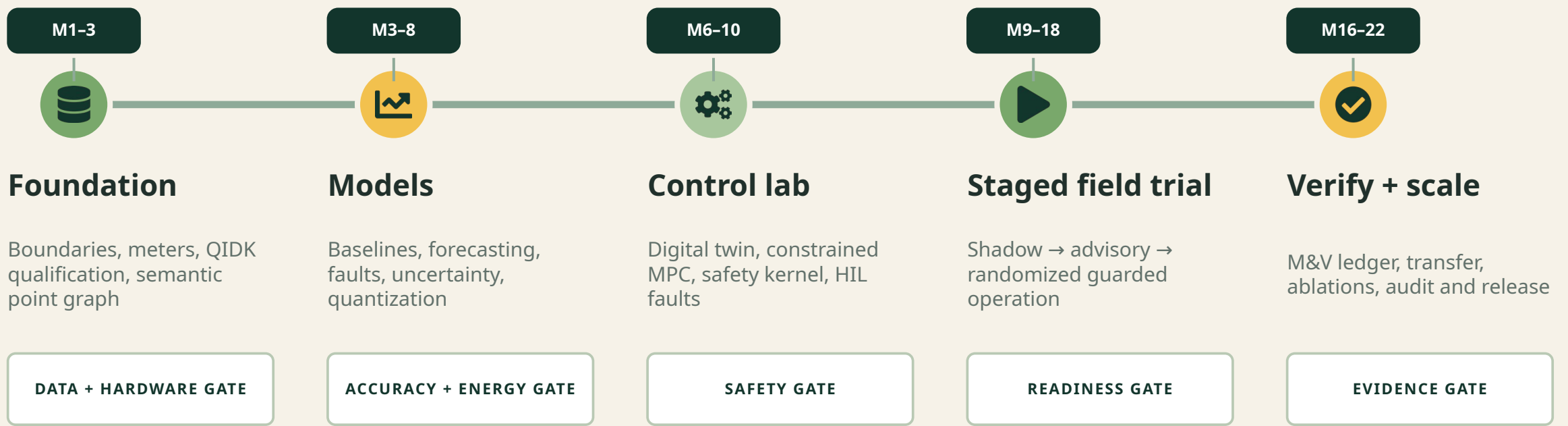
DEPLOYMENT AUTHORITY



From trustworthy data to guarded control

22 MONTHS

A 22-month evidence-gated program; work packages overlap, but authority increases only after each gate.



Success means lower adjusted energy + lower location-based carbon + non-inferior comfort/IAQ + savings greater than platform overhead